

# **Major Project: Final Year CSE Group 7**

## **Topic:**

Intelligent Web Application Firewall Using AI/ML

## **Project Description:**

This project aims to use intelligent web application firewall (WAF) that uses AI/ML algorithms to detect and prevent malicious web traffic, such as SQL injections, cross-site scripting (XSS), DDOS attack and others in real-time.

## **Core Components:**

1. Request Monitoring and Inspection Module
  - a. Capture all incoming HTTP requests and inspects critical elements such as request headers, body, URL pattern and query parameters.
  - b. Forwards relevant data to the AI/ML module for analysis.
2. AI/ML Threat Detection Module
  - a. Model Training: Trains on existing datasets containing patterns of malicious requests (SQL injections, XSS, etc.).
  - b. Request Classification: Classifies incoming requests as either legitimate or malicious in real-time.
3. Traffic Filtering and Blocking
  - a. Based on the prediction from the AI/ML module, allows or blocks HTTP requests.
  - b. Sends notification or alerts to the system admin when a threat is detected.
4. Logging and Reporting Module
  - a. Logs all requests (legitimate and malicious) along with their metadata for audit and analysis.
  - b. Offers real-time monitoring through dashboards for attack trends and statistics.
5. Dashboard and Configuration Interface
  - a. Dashboard: Display visualizations such as the number of blocked requests, most common attack types and traffic patterns.
  - b. Settings: Allow the admin to configure firewall rules and upload new attack datasets for model retraining and manage system behavior.

## 6. Database Management System

- a. Stores logs of incoming requests, blocked attacks, and configuration settings.
- b. Maintains datasets for model training and user feedback for future improvements.

### **Potential Applications:**

1. Real-Time Threat Detection and Prevention
  - a. Provides instant protection for web applications against known and unknown threats using machine learning models.
2. Enhanced Web Application Security
  - a. Protects sensitive web applications, such as e-commerce platforms, banking systems, and enterprise portals, from external threats.
  - b. Prevents data breaches, unauthorized access and data tampering by filtering out malicious traffic before it reaches the application.

### **Group Members – Group 7**

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